# JAX-sklearn (xlearn) — Transparent Acceleration Architecture

User API, compatibility layer, JAX implementations, data policy, and hardware backends

Dashed frame = architecture zone
Solid frame = subsystem
Box = module and responsibility

**PROJECT OVERVIEW**

JAX-sklearn provides a drop-in scikit-learn API while transparently accelerating supported estimators with JAX.

The architecture has a user API, a compatibility layer, JAX implementations, a data policy, and hardware backends.

## USER CODE · sklearn-compatible API

- **import xlearn as sklearn**
  100% call surface
- **Estimators**
  linear_model · cluster
  decomposition · preprocessing
- **Calls**
  fit · predict · transform

## COMPATIBILITY · xlearn/__init__.py

- **__getattr__**
  lazy submodule import
- **_auto_jax_accelerate_module**
  replace estimator classes
- **AcceleratorRegistry**
  _accelerator.py
- **create_intelligent_proxy**
  _proxy.py
- **monkey_patch_estimator**
  install proxy class

## JAX ACCELERATION · xlearn/_jax

- **EstimatorProxy**
  fit-time dispatch
- **create_accelerated_estimator**
  _accelerator.py
- **_universal_jax.py**
  JIT · vmap
- **original_class**
  sklearn fallback
- **_gp.py**
  Gaussian process
- **_mlp.py**
  neural net
- **_svm.py**
  SVM kernels
- **_tree.py**
  tree models

## DATA MANAGEMENT · arrays and policy

- **_data_conversion.py**
  NumPy ↔ JAX
  dtype · shape
- **_config.py**
  enable_jax
  jax_platform
- **jax_auto_threshold**
  complexity >= 1e8
- **detect_hardware**
  install command
  platform probe

## HARDWARE · JAX extras

- **CPU**
  jax-cpu · jaxlib
- **CUDA GPU**
  jax-gpu
  jax-cuda11
- **TPU**
  jax-tpu
- **Apple Metal**
  jax-metal
  float32 · experimental

The compatibility layer places EstimatorProxy between the sklearn API and either a JAX implementation or the original estimator.